Midterm Project- R Code and Outputs

Load in neccessary packages

```
library(tidymodels)
library(caret)
library(corrplot)
library(glmnet)
```

Part I

Using this dataset, please help Researcher A build a prediction model for PFS, aiming to understand how demographic, tumor, and treatment characteristics influence progression-free survival in breast cancer patients.

To answer this question, your report should at least include the following:

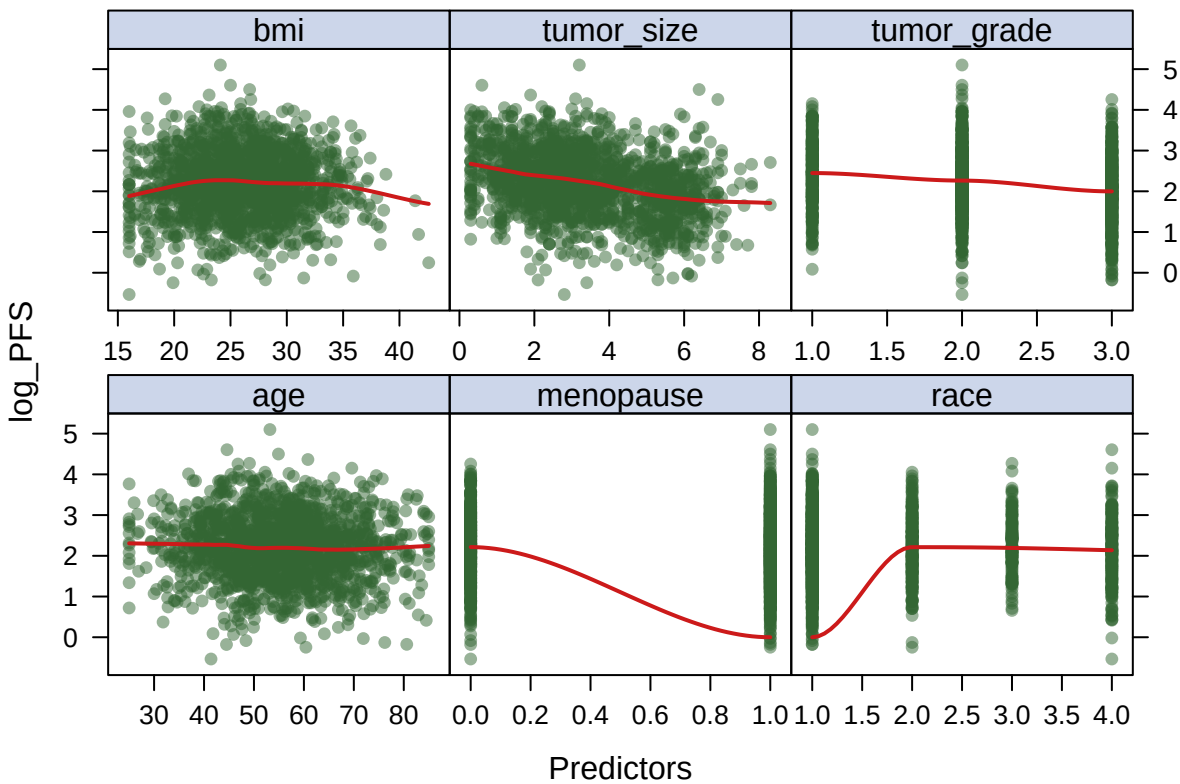
Model Training: Provide a detailed description of the model training procedure and how you obtained the model. Your description should include sufficient detail so that another statistician can reproduce the same model. Points will be deducted if the details provided are insufficient to reproduce your results.

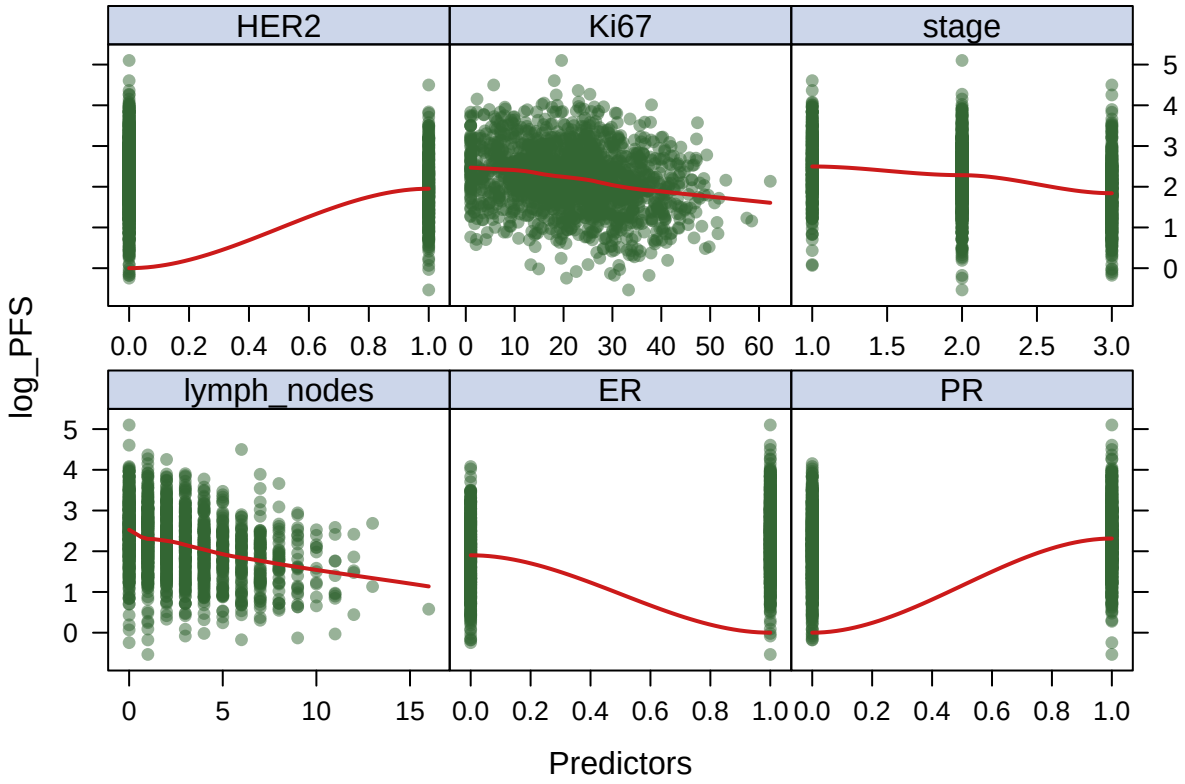
Import and partition data using 'tidymodels' package

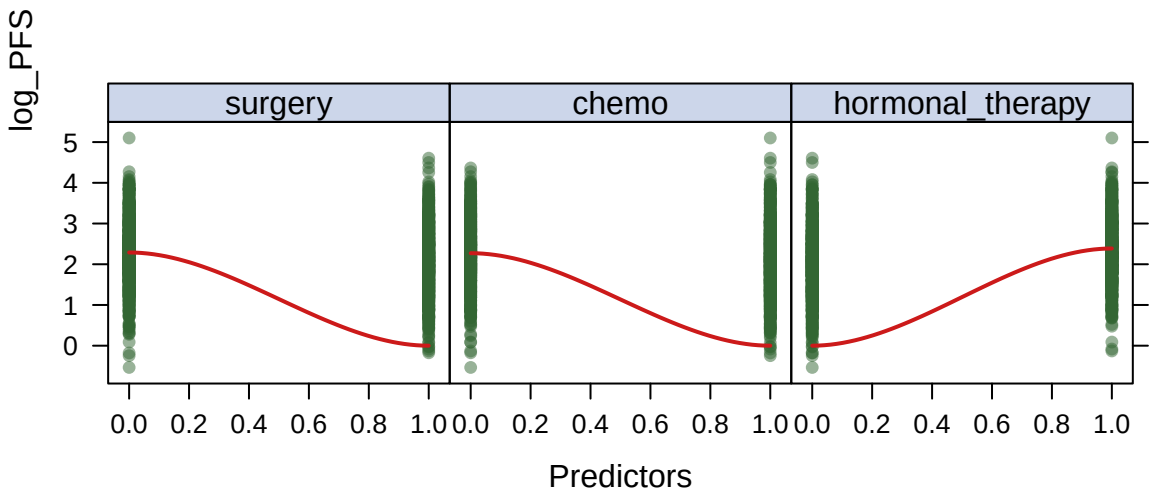
```
load("data/datA.RData")

set.seed(2222)

datA_split = initial_split(datA, prop = 0.8)

#extract the training and test data
training_datA = training(datA_split)
testing_datA = testing(datA_split)

#training data
x_trainA = model.matrix(log_PFS ~ . - id, training_datA)[, -1]
y_trainA = training_datA$log_PFS

#test data
x_testA = model.matrix(log_PFS ~ . - id, testing_datA)[, -1]
y_testA = testing_datA$log_PFS
```

Exploratory Data Analysis

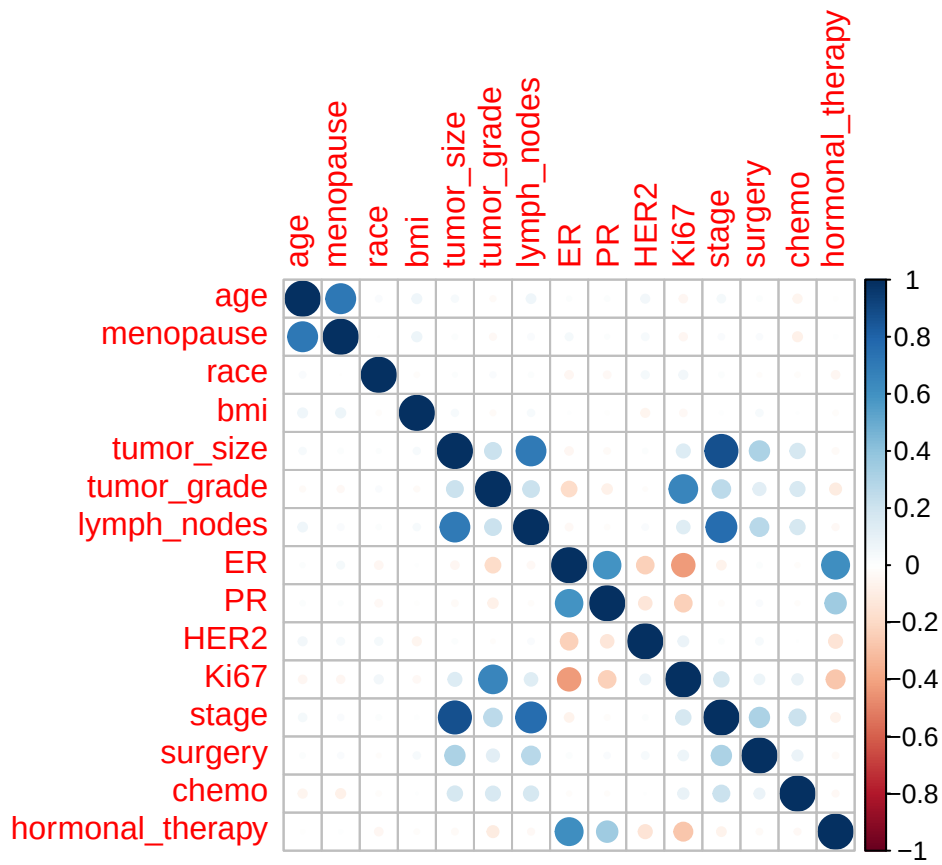
```r
# Changing the plot display
theme1 <- trellis.par.get()
theme1$plot.symbol$col <- rgb(.2, .4, .2, .5)
theme1$plot.symbol$pch <- 16
theme1$plot.line$col <- rgb(.8, .1, .1, 1)
theme1$plot.line$lwd <- 2
theme1$strip.background$col <- rgb(.0, .2, .6, .2)
trellis.par.set(theme1)

# Feature plots to look at trends between predictors and the outcome (log_PFS)
featurePlot(x = training_datA[ ,2:16],
            y = training_datA[ ,17],
            plot = "scatter",
            span = .5,
            labels = c("Predictors","log_PFS"),
            type = c("p", "smooth"),
            layout = c(3, 2))
```




HER2
Ki67
stage
lymph_nodes
ER
PR
log_PFS
Predictors




```
# Calculates the correlation matrix for all the predictors
corrplot(cor(x_trainA), method = "circle", type = "full")
```



This data set includes 7 binary predictors, 3 categorical predictors, 5 continuous predictors, and 1 continuous outcome in log scale.

Chose elastic net because continuous variables formed a cloud, meaning we could assume linearity? homoscedasticity?

Correlation plot revealed that certain variables were highly correlated. Chose to address this with elastic net because we can address colinearity for [] and [] reasons

```
# Elastic Net

ctrl1 <- trainControl(method = "cv", number = 10) # set up cv
set.seed(2) # set seed for reproduction

enet.fit <- train(log_PFS ~ . - id,
                  data = training_datA,
                  method = "glmnet",
                  tuneGrid = expand.grid(alpha = seq(0, 1, length = 21),
                                         lambda = exp(seq(-1, -7, length = 100))),
                  trControl = ctrl1)
```

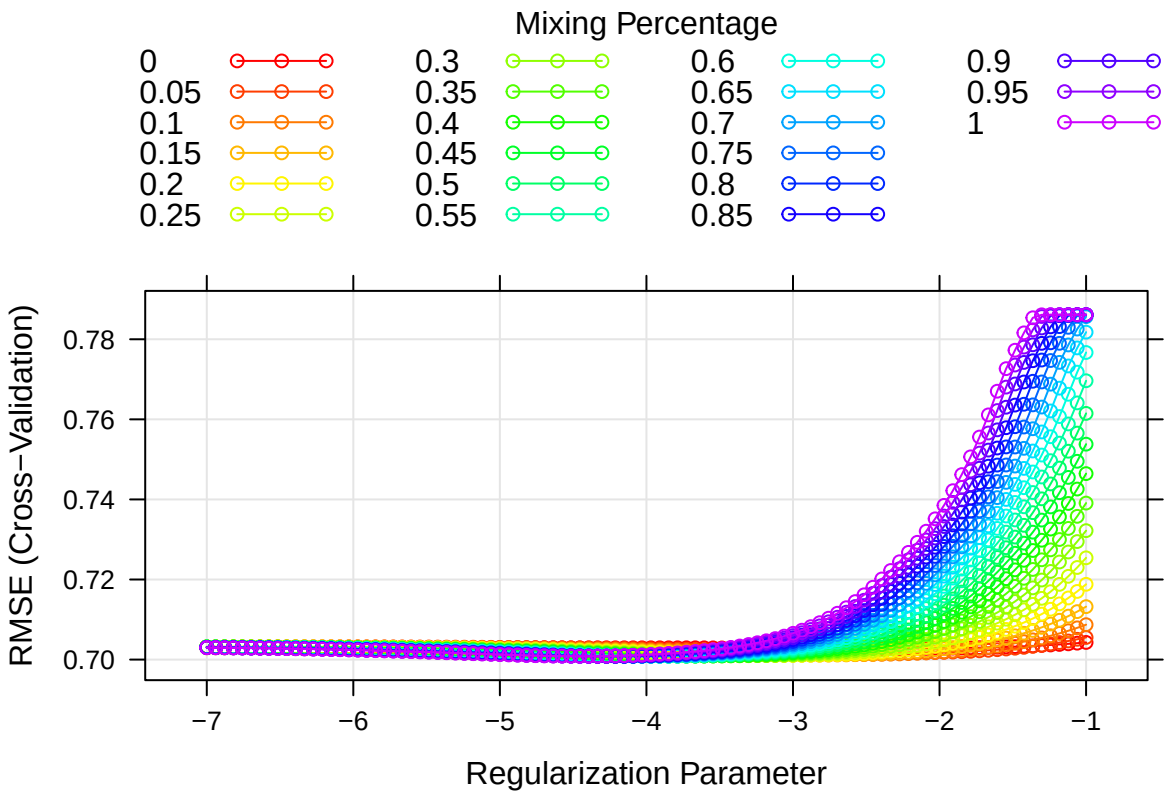
```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```
enet.fit$bestTune
```

```
##      alpha     lambda
```

```
## 1944  0.95 0.01235197
```

```r
myCol <- rainbow(25)
myPar <- list(superpose.symbol = list(col = myCol),
superpose.line = list(col = myCol))

plot(enet.fit, par.settings = myPar, xTrans = log)
```




```r
# coefficients in the final model
coef(enet.fit$finalModel, enet.fit$bestTune$lambda)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##                   s=0.01235197
## (Intercept)        2.656050011
## age               -0.001046560
## menopause          .
## race               .
## bmi                .
## tumor_size        -0.057703954
## tumor_grade       -0.066041243
## lymph_nodes       -0.056993530
## ER                 0.220684085
## PR                 .
## HER2              -0.176250264
## Ki67              -0.003932773
```

```
## stage             .
## surgery           .
## chemo             .
## hormonal_therapy  0.137999051
```

Results: Report the model. Interpret the model and assess its performance.

Part II

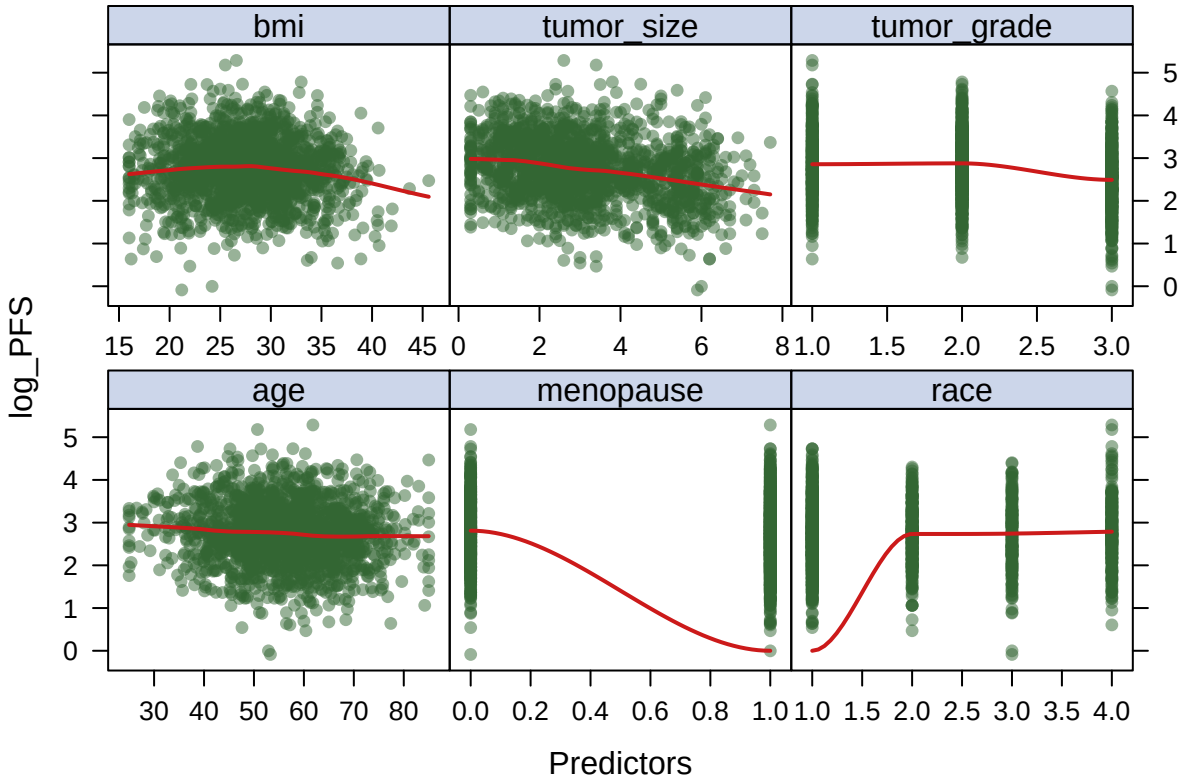
About a decade later, Researcher B conducted a separate study enrolling 2000 breast cancer patients diagnosed between 2000 and 2005, using the same study design and variable definitions. Like Researcher A's study, all patients were followed until disease progression or death. Researcher B's data are stored in datB.RData. Researcher A would like to evaluate whether the prediction model built on the original data generalizes well to Researcher B's dataset. Does the model generalize well? If yes, provide justification. If not, build a new prediction model for the 2000–2005 cohort and discuss what has changed in prediction.

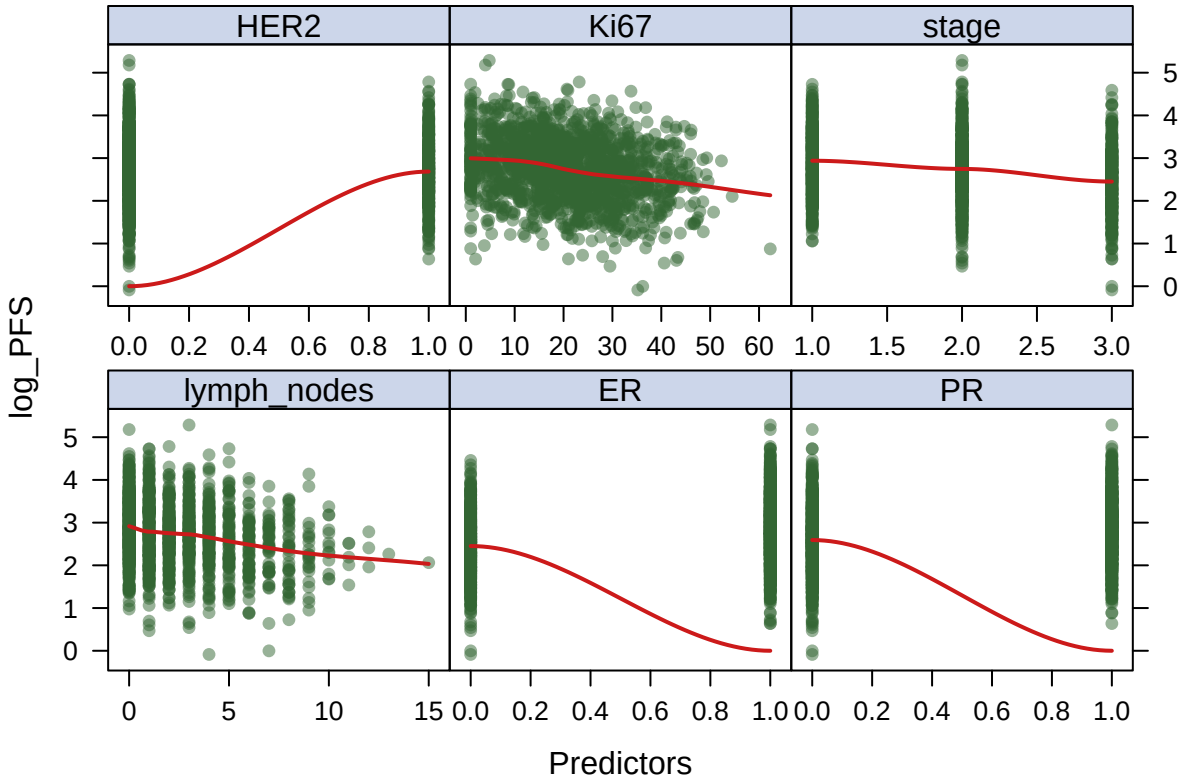
Import and partition data using 'tidymodels' package

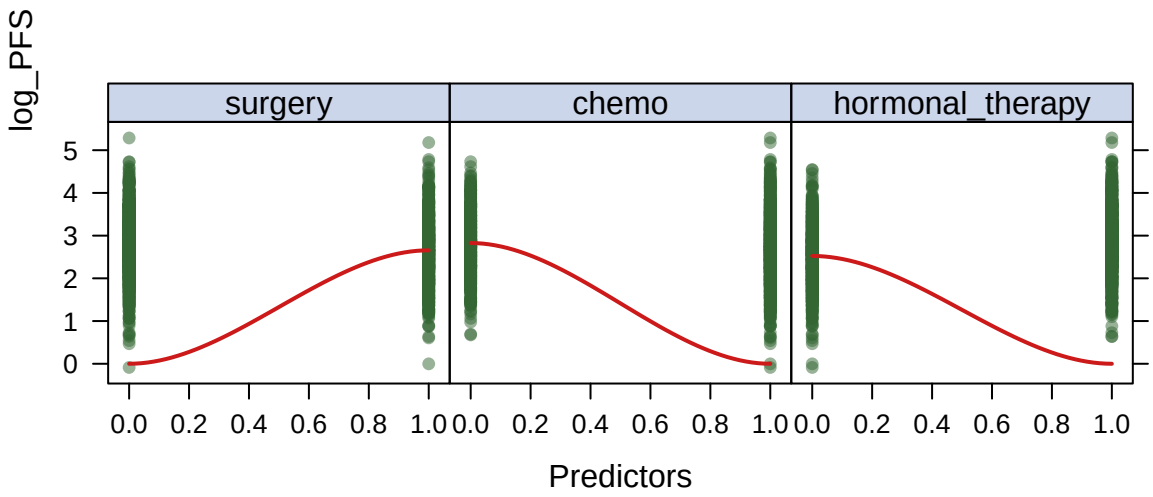
```r
load("data/datB.RData")

set.seed(2222)

datB_split = initial_split(datB, prop = 0.8)

#extract the training and test data
training_datB = training(datB_split)
testing_datB = testing(datB_split)

#training data
x_trainB = model.matrix(log_PFS ~ . - id, training_datB)[, -1]
y_trainB = training_datB$log_PFS

#test data
x_testB  = model.matrix(log_PFS ~ . - id, testing_datB)[, -1]
y_testB = testing_datB$log_PFS
```

Exploratory Data Analysis

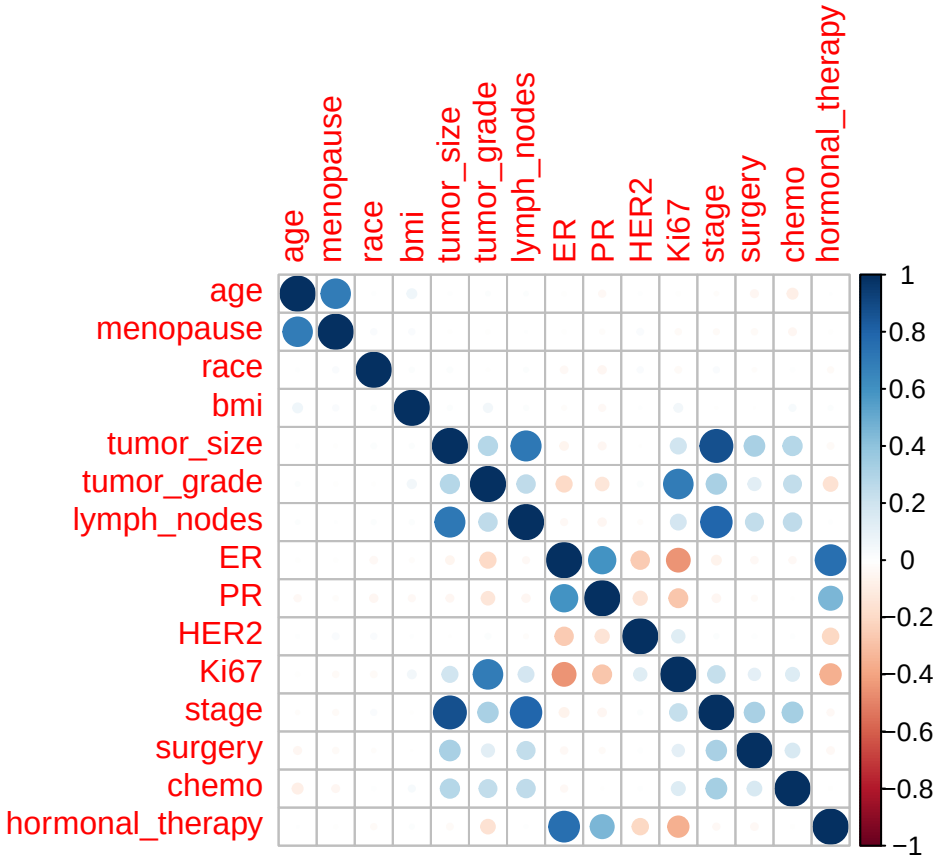
```r
# Feature plots to look at trends between predictors and the outcome (log_PFS)
featurePlot(x = training_datB[ ,2:16],
            y = training_datB[ ,17],
            plot = "scatter",
            span = .5,
            labels = c("Predictors","log_PFS"),
            type = c("p", "smooth"),
            layout = c(3, 2))
```


bmi
tumor_size
tumor_grade
age
menopause
race
log_PFS
Predictors
15 20 25 30 35 40 45
0 2 4 6 8
1.0 1.5 2.0 2.5 3.0
30 40 50 60 70 80
0.0 0.2 0.4 0.6 0.8 1.0
1.0 1.5 2.0 2.5 3.0 3.5 4.0
0 1 2 3 4 5


HER2
Ki67
stage
lymph_nodes
ER
PR
log_PFS
Predictors
0.0 0.2 0.4 0.6 0.8 1.0
0 10 20 30 40 50 60
1.0 1.5 2.0 2.5 3.0
0 5 10 15
0 1 2 3 4 5




```
# Calculates the correlation matrix for all the predictors
corrplot(cor(x_trainB), method = "circle", type = "full")
```



Run GAM or MARs if model isn't super generalizable?